

Radicals Do Not Split Across a Sum

Testing a suspect rule with numbers, splitting radicals over products and quotients only, and finding and fixing the split-over-a-sum error

Directions

- A radical comes apart over a *product* and a *quotient*: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ and $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$ for $a, b \geq 0$.
 - There is **no** matching rule for a sum. Whenever you are tempted to write $\sqrt{a+b}$ as $\sqrt{a} + \sqrt{b}$, test it with two numbers first.
 - The same warning applies in exponent notation: $(a+b)^{1/2}$ is not $a^{1/2} + b^{1/2}$.
 - Four problems show work a student handed in. Compute the correct values first, then say what went wrong.
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1. Work with the numbers 9 and 16.

(a) Evaluate $\sqrt{9+16}$: _____ (b) Evaluate $\sqrt{9} + \sqrt{16}$: _____

(c) Explain what the comparison of those two values tells you about taking a square root term by term across a sum.

2. Write each radical in simplest form.

(a) $\sqrt{50} =$ _____ (b) $\sqrt{72} =$ _____

3. A student wrote: “ $\sqrt{x^2 + 9} = x + 3$, because the root goes onto each piece.”

(a) Evaluate the left side, $\sqrt{x^2 + 9}$, at $x = 4$: _____

(b) Evaluate the right side, $x + 3$, at $x = 4$: _____

(c) Name the error, and say what the simplest correct form of $\sqrt{x^2 + 9}$ is.

4. Work in exponent notation with the numbers 36 and 64.

(a) Evaluate $(36 + 64)^{1/2}$: _____ (b) Evaluate $36^{1/2} + 64^{1/2}$: _____

5. Write each expression in simplest form.

(a) $\sqrt{75} =$ _____ (b) $\sqrt{\frac{4}{9}} =$ _____

6. A student simplified $\sqrt{4 + x^2}$ as $2 + x$.

(a) Evaluate $\sqrt{4 + x^2}$ at $x = 3$, rounded to two decimal places: _____

(b) Evaluate the student's expression $2 + x$ at $x = 3$: _____

(c) Name the error, and give the simplest correct form of $\sqrt{4 + x^2}$.

7. A student solved $\sqrt{x} + \sqrt{9} = 5$ like this:

“ $\sqrt{x + 9} = 5$, so $x + 9 = 25$, so $x = 16$.”

(a) Solve the original equation correctly: _____

(b) Name the error in the first line of the student's work.

8. Challenge. Consider $\sqrt{12} + \sqrt{27}$.

(a) Write it as a single simplified radical (exact form): _____

(b) Give that value as a decimal, rounded to two places: _____

(c) Give $\sqrt{12 + 27}$ as a decimal, rounded to two places: _____